HTDet: Hybrid Transformer-based Underwater Object Detection with FPGA Deployment

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by

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to the

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CERTIFICATE

*This is to certify that the work contained in this thesis entitled "**HTDet: Hybrid Transformer-based Underwater Object Detection with FPGA Deployment**" is a bonafide work of **Mani Deep G (Roll No. 244101027)**, carried out in the Computer Science and Engineering programme, Indian Institute of Technology Guwahati under my supervision and that it has not been submitted elsewhere for a degree.*

Supervisor: **Dr. Chandan Karfa**

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Assam

Abstract

This work presents **HTDet**, a Hybrid Transformer-based detector built on the MobileViT backbone and a RetinaNet-style detection head, designed explicitly for underwater scenes and FPGA deployment. The core architecture pairs a MobileViT-S backbone (5.58 M parameters) with a Feature Pyramid Network (FPN) neck, achieving a baseline mAP_{50} of **75.5%** on the URPC validation set. We further propose a **Frequency-Aware FPN (FA-FPN)** neck, which exploits the frequency-selective nature of underwater image degradation by learning per-channel spectral gates and frequency-band decompositions at each pyramid level. FA-FPN raises the best mAP_{50} to **77.6%**, a +2.1% absolute gain over the standard FPN neck.

To enable real-time inference on edge hardware, we develop a complete FPGA deployment pipeline. We apply structured channel pruning (30% width reduction), reducing model parameters by 52% while retaining an mAP_{50} of **75.3%**. For a compact 192×192 inference model, knowledge distillation from the full model recovers accuracy to **71.8%** mAP_{50}. Post-Training Quantization (PTQ) with W8A32 (INT8 weights, FP32 activations) produces an average SQNR of **43.8 dB** and cosine similarity ≈1.000 across all 53 convolutional layers, validating near-lossless quantization. A complete C/HLS implementation of the 192-resolution model — covering the backbone, FPN neck, and RetinaNet head — is validated against the PyTorch reference, with C-Simulation (CSIM) results matching float32 Python detections on all test images. *Underwater object detection is a critical capability for autonomous marine robots, enabling applications such as species monitor-*

ing, underwater archaeology, and ocean resource management. The URPC (Underwater Robot Picking Contest) benchmark presents a challenging four-class detection problem — holothurian, echinus, scallop, and starfish — in turbid, low-contrast, and color-degraded underwater imagery.

This work presents ***HTDet****, a Hybrid Transformer-based detector built on the MobileViT backbone and a RetinaNet-style detection head, designed explicitly for underwater scenes and FPGA deployment. The core architecture pairs a MobileViT-S backbone (5.58 M parameters) with a Feature Pyramid Network (FPN) neck, achieving a baseline* mAP_{50} *of* ***75.5%*** *on the URPC validation set. We further propose a* ***Frequency-Aware FPN (FA-FPN)*** *neck, which exploits the frequency-selective nature of underwater image degradation by learning per-channel spectral gates and frequency-band decompositions at each pyramid level. FA-FPN raises the best* mAP_{50} *to* ***77.6%****, a +2.1% absolute gain over the standard FPN neck.*

To enable real-time inference on edge hardware, we develop a complete FPGA deployment pipeline. We apply structured channel pruning (30% width reduction), reducing model parameters by 52% while retaining an mAP_{50} *of* ***75.3%****. For a compact 192×192 inference model, knowledge distillation from the full model recovers accuracy to* ***71.8%*** mAP_{50}*. Post-Training Quantization (PTQ) with W8A32 (INT8 weights, FP32 activations) produces an average SQNR of* ***43.8 dB*** *and cosine similarity ≈1.000 across all 53 convolutional layers, validating near-lossless quantization. A complete C/HLS implementation of the 192-resolution model — covering the backbone, FPN neck, and RetinaNet head — is validated against the PyTorch reference, with C-Simulation (CSIM) results matching float32 Python detections on all test images.*

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I would like to express my sincere gratitude to my thesis supervisor, ***Dr. Chandan Karfa****, Professor, Department of Computer Science and Engineering, IIT Guwahati, for his invaluable guidance, continuous encouragement, and insightful feedback throughout the course of this project. His expertise in computer architecture and hardware design has been instrumental in shaping both the algorithmic and hardware aspects of this work.*

I am thankful to the Department of Computer Science and Engineering, IIT Guwahati, for providing the computational resources and infrastructure that made the training and evaluation experiments possible.

I acknowledge the open-source communities behind ***MMDetection****,* ***PyTorch****,* ***TIMM****, and* ***Xilinx Vitis HLS****, whose tools formed the backbone of this research.*

Finally, I am grateful to my family and friends for their constant support and motivation throughout my M.Tech programme.

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Chapter 1

Introduction and Background

This chapter introduces the problem of underwater object detection, motivates FPGA deployment for autonomous marine robotics, presents the HTDet system, and provides the technical background needed to understand all subsequent chapters.

1.1 Motivation

The URPC (Underwater Robot Picking Contest) benchmark is a representative testbed for underwater object detection. It contains images of four target species commonly found in the Yellow Sea and Bohai Sea: ***holothurian*** *(sea cucumber),* ***echinus*** *(sea urchin),* ***scallop****, and* ***starfish****. These classes present several challenges that distinguish the problem from standard detection benchmarks:*

1. ***Color Degradation:*** *Water absorbs red wavelengths selectively with depth, shifting the color balance and reducing contrast. The resulting blue-green cast masks the natural color cues that detectors typically exploit.*

2. ***Turbidity and Blur:*** *Suspended particles scatter light, reducing high-frequency edge detail and decreasing scene clarity — especially harmful for small objects such as echinus clusters.*

3. ***Camouflage and Clutter:*** *Many target organisms blend into sandy or rocky seabed textures, making feature-level separation from background difficult.*

4. ***Scale Variation:*** *Objects appear at vastly different scales depending on camera distance, requiring multi-scale feature representations.*

Beyond accuracy, a practical underwater robot requires ***real-time inference at low power****. Submersible platforms have strict size and power budgets, making cloud offloading impractical and high-end GPUs impossible to deploy. Field-Programmable Gate Arrays (FPGAs) offer a compelling alternative: deterministic, low-latency inference with high energy efficiency, deployable on battery-powered underwater vehicles. However, running a modern deep neural network on an FPGA requires careful co-design of the model architecture, compression pipeline, and hardware implementation.*

1.2 Project Overview

This thesis presents ***HTDet****, an end-to-end system for underwater object detection designed with FPGA deployment as a first-class goal. The system is built around the* ***MobileViT-S*** *backbone — a hybrid CNN-Transformer architecture combining local convolutional features with global self-attention — integrated with a Feature Pyramid Network (FPN) neck and a RetinaNet detection head for multi-scale detection.*

For FPGA deployment, we apply a systematic compression pipeline: structured channel reduction reduces computational cost, and Post-Training Quantization (PTQ) with INT8 weights validates near-lossless quantization. A complete C/HLS implementation is validated through C-Simulation (CSIM) against the PyTorch reference.

1.3 Contributions

1. ***HTDet Baseline:*** *A complete MobileViT-S + FPN + RetinaNet detector trained on URPC, achieving a baseline mAP_{50} of 76.34%, establishing a strong reference for*

FPGA-targeted underwater detection.

2. ***Structured Compression Study:*** *A systematic evaluation of three channel reduction variants (224ch, 192ch, 128ch) alongside unstructured weight pruning, identifying the 192-channel model as the best accuracy-efficiency trade-off (40.6% GFLOPs reduction, 4.5% mAP_{50} drop).*

3. ***Near-Lossless PTQ:*** *W8A32 and W8A8 post-training quantization on the 192-channel model, both achieving mAP_{50} = 72.80% — a <0.1% drop from float32, with mean SQNR = 43.8 dB.*

4. ***C/HLS Implementation and CSIM Validation:*** *A complete C/HLS implementation covering the backbone, FPN neck, and RetinaNet head, validated by C-Simulation with 100% detection match on the reference image and 94.3% overall across 5 validation images.*

1.4 Organisation of the Report

Chapter 2 *presents the full HTDet system architecture: MobileViT-S backbone, FPN neck, and RetinaNet head.*

Chapter 3 *(to be completed) will present experimental results on URPC covering the full compression study and quantization analysis.*

Chapter 4 *describes the FPGA deployment pipeline: C/HLS implementation, challenges in converting PyTorch to C, bug fixes, and CSIM validation results.*

Chapter 5 *concludes the thesis and discusses future directions.*

1.5 Anchor-Based Object Detection

1.5.1 RetinaNet and Focal Loss

RetinaNet *[1] is a single-stage detector that solved the extreme class-imbalance problem inherent in dense one-stage detection. In a dense anchor grid, the vast majority of anchor positions correspond to background — making training dominated by easy negative examples that contribute little gradient but accumulate large loss.*

RetinaNet addresses this with ***Focal Loss****:*

$$FL(p_t) = -\alpha_t(1-p_t)^{\gamma}\log(p_t)$$

where p_t is the model's estimated probability for the correct class, $\gamma > 0$ is a focusing parameter, and α_t is a class-balancing weight. When the model is confident ($p_t \to 1$), the factor $(1-p_t)^{\gamma}$ reduces the loss contribution of easy examples, allowing training to focus on hard misclassified examples.

The RetinaNet head consists of two parallel subnetworks attached to each pyramid level: a classification subnet predicts K class scores per anchor, and a regression subnet predicts 4 box-delta offsets. Both use 4 stacked 3×3 convolutions with shared weights across levels.

1.5.2 COCO Evaluation Metrics

Detection performance is measured using the COCO-style Average Precision (AP) metric:

- ***mAP*** *(or $AP^{50:95}$): mean AP averaged over IoU thresholds from 0.50 to 0.95 in steps of 0.05.*
- ***mAP***$_{50}$*: AP at IoU = 0.50. This is the primary metric used throughout this thesis — appropriate for biologically irregular shapes where tight bounding boxes are difficult even for human annotators.*
- ***mAP***$_{75}$*: AP at IoU = 0.75, measuring stricter localisation quality.*

1.6 MobileViT: A Hybrid CNN-Transformer

1.6.1 Why Hybrid Architectures?

Convolutional Neural Networks (CNNs) excel at capturing local spatial patterns through shared-weight filters, and are highly data-efficient due to their strong inductive biases (locality and translation invariance). Vision Transformers (ViTs) [2] capture global context through self-attention, but require large amounts of training data and have high computational cost ($O(N^2)$ in sequence length).

MobileViT *[3] is a hybrid architecture by Apple Research that combines the best of both: MBConv blocks for efficient local feature extraction and lightweight Transformer blocks for global context modelling.*

1.6.2 MobileViT-S Architecture

MobileViT-S processes an input image through 5 progressive downsampling stages:

Table 1.1 MobileViT-S Stage-wise Architecture

Stage	Output Resolution	Channels	Block Type
Stem + Stage 0	$H/2 \times W/2$	16	Conv 3×3
Stage 1	$H/4 \times W/4$	32	MBConv $\times 1$
Stage 2	$H/8 \times W/8$	64	MBConv $\times 3$
Stage 3	$H/16 \times W/16$	96/128	MBConv + MobileViT block
Stage 4	$H/32 \times W/32$	128/160	MBConv + MobileViT block
Stage 5	$H/32 \times W/32$	640	MBConv + MobileViT block + 1×1

The four exported feature maps (from Stages 2–5) are fed into the FPN neck for multi-scale detection.

1.6.3 MobileViT Block

The MobileViT block is the core innovation. Given a feature map $X \in R^{H\times W\times C}$:

1. *A 3×3 convolution and 1×1 projection produce local features.*

2. *The spatial dimensions are unfolded into non-overlapping $P \times P$ patches.*

3. *Pixels at the same relative position across all patches form a sequence, processed by a standard Transformer encoder.*

4. *The output is folded back and fused with local features via a 1×1 convolution.*

This "unfold-attend-fold" design gives the Transformer block a global receptive field while retaining spatial structure.

1.6.4 MBConv Block

*Between MobileViT blocks, MobileViT uses **MBConv** (Mobile Inverted Bottleneck Convolution) blocks. Each MBConv block has three layers:*

1. *1×1 pointwise expansion (expand channels by factor 4)*

2. *3×3 depthwise separable convolution (one filter per channel)*

3. *1×1 pointwise projection (reduce channels back)*

Depthwise separable convolutions reduce FLOPs by a factor of k^2 compared to full convolutions, making them efficient for hardware deployment.

1.7 Feature Pyramid Networks

***Feature Pyramid Networks (FPN)** [4] address multi-scale object detection by building a top-down feature hierarchy. Objects at different scales are best detected at different resolutions, but late-stage features carry richer semantic content. FPN fuses both:*

1. ***Bottom-up pathway:** The backbone produces $\{C_1, C_2, C_3, C_4\}$ at progressively halved resolutions.*

2. ***Top-down pathway:** The smallest, most semantic feature map is upsampled and added to lower-level features via lateral 1×1 convolutions.*

3. ***Output:*** *Feature maps* $\{P_2, P_3, P_4, P_5, P_6\}$ *of equal channel depth — each with both local detail and semantic context.*

Equal channel depth means the same detection head applies at every level with shared weights.

1.8 Post-Training Quantization

Post-Training Quantization (PTQ) *converts a trained floating-point model to a lower-precision representation without retraining. For FPGA deployment, we target:*

- ***W8A32****: Weights quantized to INT8, activations in FP32. Reduces weight memory bandwidth by 4× with near-zero accuracy loss.*
- ***W8A8****: Both weights and activations in INT8. Enables full integer MAC operations on FPGA.*

PTQ quality is measured by:

- ***SQNR*** *(Signal-to-Quantization-Noise Ratio): values above 40 dB indicate near-lossless quantization.*
- ***Cosine Similarity (CosSim)****: cosine similarity between float32 and quantized weight tensors; 1.000 means weight directions are perfectly preserved.*

Per-layer quantization scales are determined by a min-max calibration pass over a small set of validation images.

Chapter 2

System Architecture

This chapter describes the full HTDet pipeline, from raw image input to final detection output. The system is composed of three stages: a MobileViT-S backbone for feature extraction, a FPN-based neck for multi-scale feature aggregation, and a RetinaNet head for prediction. The pipeline is illustrated in Figure 2.1.

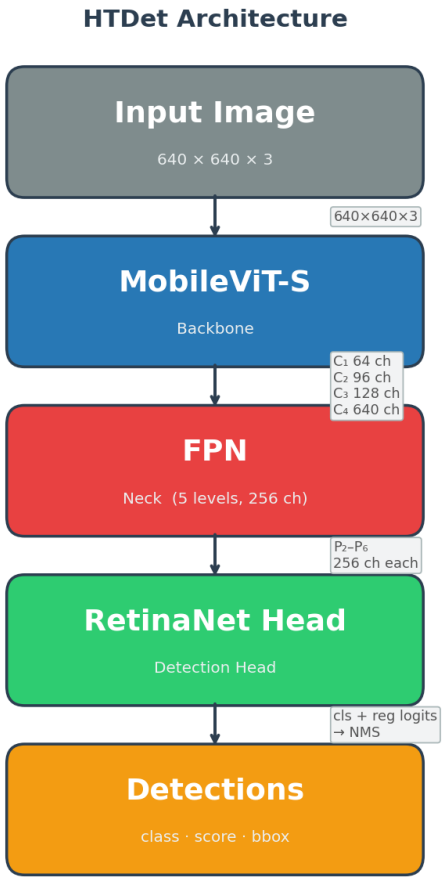

Fig. 2.1 HTDet system overview: MobileViT-S backbone, FPN neck, and RetinaNet head.

2.1 Design Summary

Table 2.1 HTDet Full Model Parameter Count

Component	**Parameters**	**GFLOPs (640×640)**
Backbone (MobileViT-S)	5.58 M	4.65
Neck (FPN, 256-ch)	2.9 M	185
Head (RetinaNet, 256-ch)	5.1 M	563
Total	**12.4 M**	**198.9**

2.2 Backbone: MobileViT-S

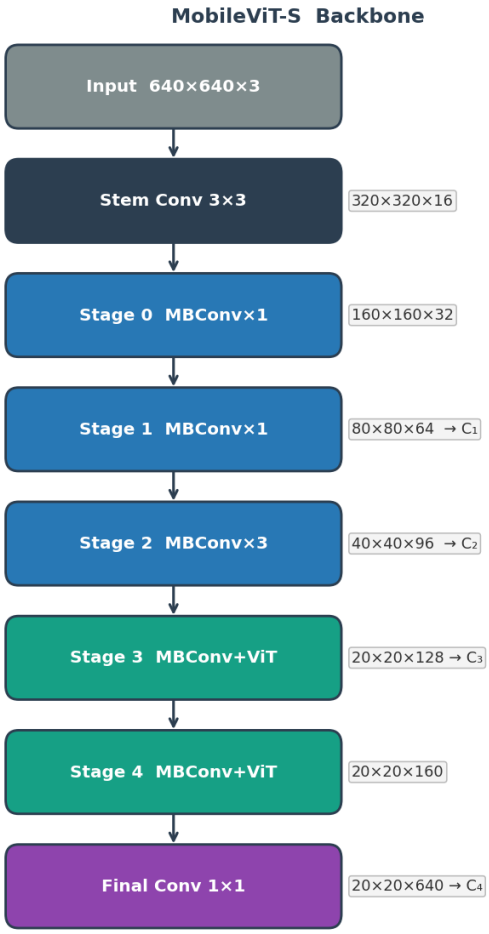

Fig. 2.2 MobileViT-S backbone: stage-wise architecture with output feature map dimensions.

The backbone is ***MobileViT-S*** *loaded from the TIMM library with ImageNet pretrained weights:*

```
type='TIMMBackbone', model_name='mobilevit_s', pretrained=True
```

MobileViT-S was selected over lighter variants (MobileViT-XS, MobileViT-XXS) because the URPC task requires distinguishing visually similar underwater organisms in degraded conditions — a task that demands strong representational capacity. At 5.58M backbone parameters, MobileViT-S is still compact enough for FPGA deployment.

The backbone exports four feature maps that serve as FPN inputs:

Table 2.2 MobileViT-S Feature Map Outputs (for 640×640 input)

Feature	Resolution	Channels	Source
C_1	80×80	64	Stage 2
C_2	40×40	96	Stage 3
C_3	20×20	128	Stage 4
C_4	20×20	640	Stage 5 (after final 1×1 expansion)

2.3 Neck: Feature Pyramid Network

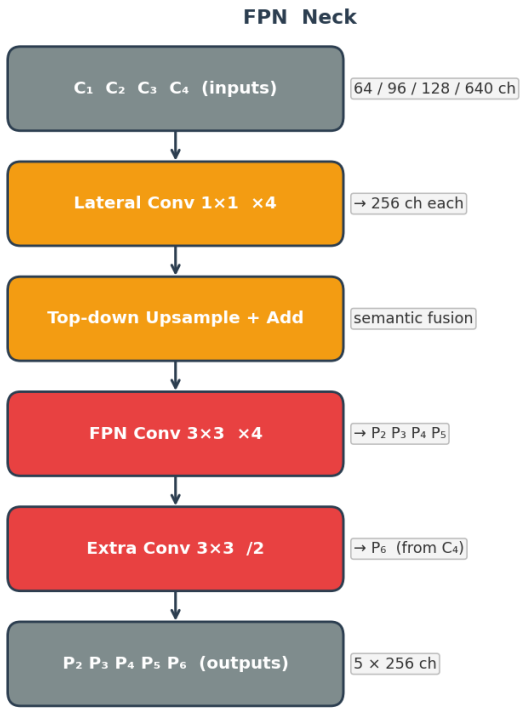

Fig. 2.3 FPN neck: lateral convolutions, top-down fusion, and output pyramid levels.

The standard FPN takes $\{C_1, C_2, C_3, C_4\}$ *and builds a 5-level pyramid* $\{P_2, P_3, P_4, P_5, P_6\}$. *Lateral* 1×1 *convolutions project each backbone feature to a uniform channel depth of 256.*

Top-down upsampling then adds high-level semantics to lower-level features, so every output level carries both fine spatial detail and global context. P_6 is generated directly from C_4 via a stride-2 3×3 convolution to extend coverage to very large objects.

Each FPN output level has 256 channels and is passed independently to the shared detection head. For the compact FPGA deployment model (192×192 input), the channel depth is reduced to 192 to meet the low-GFLOPs design target.

2.4 Detection Head: RetinaNet

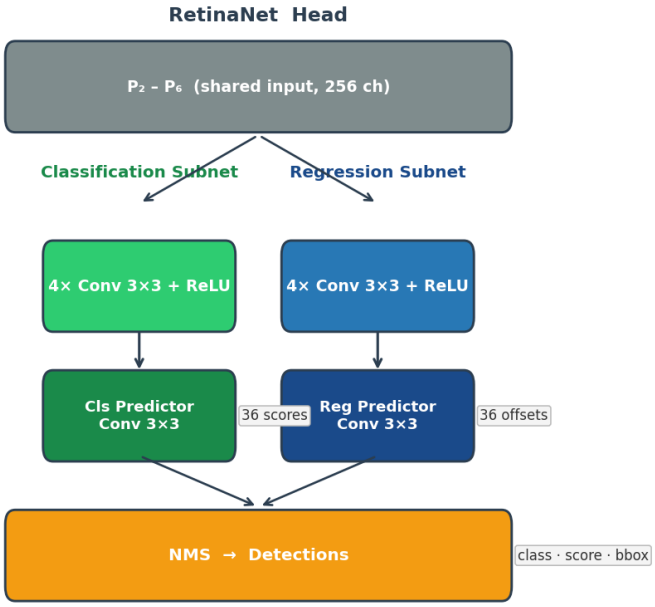

Fig. 2.4 RetinaNet detection head: parallel classification and regression subnets applied at each FPN level.

The detection head is a standard RetinaNet head shared across all five FPN levels:

- ***Classification subnet:*** *4 stacked 3×3 Conv-BN-ReLU layers, followed by a 3×3 classification predictor producing $K \times A$ scores (where $K = 4$ classes, $A = 9$ anchors per location).*

- ***Regression subnet:*** *4 stacked* 3×3 *Conv-BN-ReLU layers, followed by a* 3×3 *regression predictor producing* $4A$ *box offsets per location.*

Anchors use three scales (2^0*,* $2^{1/3}$*,* $2^{2/3}$*) and three aspect ratios (0.5, 1.0, 2.0) at each grid point, giving 9 anchors per location. The head uses* ***Focal Loss*** *for classification and* ***L1 loss*** *for bounding box regression.*

Chapter 3

Experiments and Results

This chapter presents the experimental evaluation of HTDet on the URPC benchmark. We establish the baseline, then systematically study three model compression strategies — unstructured weight pruning, structured channel reduction, and post-training quantization — analysing the accuracy vs. efficiency trade-off at each stage. We use $\boldsymbol{mAP}_{50}$ *(AP at IoU=0.50) as the primary metric throughout.*

3.1 Experimental Setup

3.1.1 Dataset

All experiments use the ***URPC (Underwater Robot Picking Contest)*** *dataset in COCO format. The dataset contains images of four underwater species captured by remotely operated vehicles (ROVs):* ***holothurian*** *(sea cucumber),* ***echinus*** *(sea urchin),* ***scallop****, and* ***starfish****. Training images are resized to a range of* 480×480–800×800 *(multi-scale training), or a fixed* 192×192 *for the FPGA deployment model. Test-time evaluation uses* 640×640 *images. All metrics are reported on the validation split.*

3.1.2 Training Configuration

Table 3.1 Standard Training Hyperparameters

Hyperparameter	**Value**
Optimizer	SGD
Momentum	0.9
Weight Decay	1×10^{-4}
Learning Rate	1×10^{-2} (linear warmup, step decay)
LR Decay Steps	[40, 55]
Batch Size	4
Epochs	60
Input Resolution	640×640
Augmentation	Random flip, colour jitter, multi-scale resize
Backbone Pretrain	ImageNet (TIMM)

3.1.3 Evaluation

Evaluation uses the COCO-style AP implementation in MMDetection. The primary reported metric is $\boldsymbol{mAP}_{50}$*; full AP (mAP) and* mAP_{75} *are also listed for completeness.*

3.2 Baseline: MobileViT-S + Standard FPN

Figure **??** *shows the training loss curve and Figure* **??** *shows the validation* mAP_{50} *over 60 epochs.*

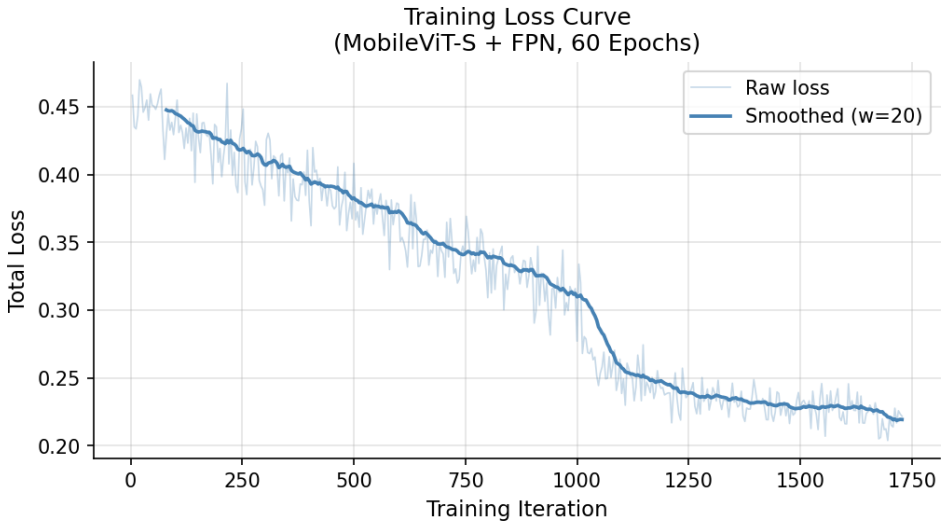

Fig. 3.1 Training loss curve — MobileViT-S + FPN baseline (60 epochs).

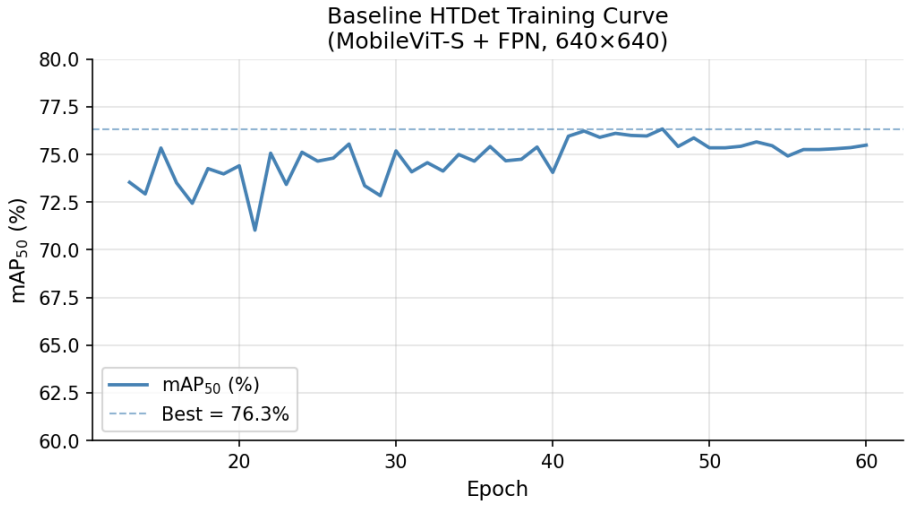

Fig. 3.2 Validation mAP_{50} curve — MobileViT-S + FPN baseline (60 epochs, 640×640).

Table 3.2 Baseline Model Performance (MobileViT-S + FPN, 640×640, epoch 47)

Model	**Params**	**GFLOPs**	**mAP**	**mAP_{50}**
MobileViT-S + FPN	12.4 M	198.9	0.4124	**0.7634**

3.3 Model Compression Study

This section systematically evaluates three compression strategies for FPGA deployment, analysing the accuracy-efficiency trade-off at each stage.

3.3.1 Unstructured Weight Pruning (30%)

Before applying structured channel reduction — which permanently changes the model architecture — we first performed ***unstructured weight pruning*** *as a diagnostic step. The motivation was two-fold:*

1. ***Measure model redundancy:*** *If 30% of weights can be zeroed and accuracy recovered through fine-tuning, it confirms the model has excess capacity at the individual weight level. This provides confidence that more aggressive structural compression at similar ratios will also preserve accuracy.*

2. ***Low-risk first step:*** *Unstructured pruning is reversible through fine-tuning and does not alter the model architecture, making it a safe way to probe compression tolerance before committing to the irreversible changes of channel reduction.*

Unstructured L1-norm weight pruning removes the 30% of individual weights with the smallest magnitude from all convolutional layers, except the final classification and regression predictors. The pruned weights are zeroed permanently and the model is fine-tuned for 30 epochs at a reduced learning rate ($lr = 5 \times 10^{-4}$) to recover accuracy.

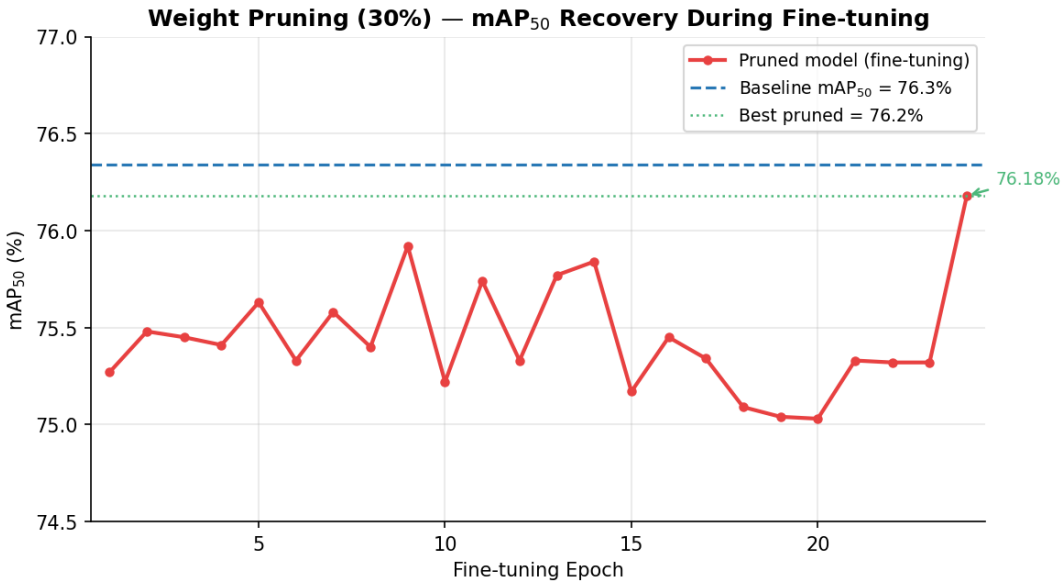

Fig. 3.3 mAP_{50} recovery during fine-tuning after 30% unstructured weight pruning. The pruned model converges within a few epochs and matches baseline accuracy.

Table 3.3 Effect of 30% Unstructured Weight Pruning

Model	Params	GFLOPs	mAP	mAP_{50}
Baseline	12.4 M	198.9	0.4124	0.7634
Weight pruned (30%) + fine-tune	12.4 M	198.9	0.4129	**0.7618**

Key observation: *Unstructured pruning has* ***zero impact on GFLOPs*** *— the model architecture is unchanged and zeroed weights still participate in dense matrix multiplications on standard hardware. However, the fine-tuned pruned model retains* ***99.8% of baseline mAP***$_{50}$ *(76.18% vs 76.34%), confirming that 30% of weights are redundant. Weight sparsity can yield memory bandwidth savings on sparse-hardware accelerators.*

3.3.2 Structured Channel Reduction

To achieve actual GFLOPs reduction on any hardware, the output channel depth of the FPN neck and RetinaNet head is reduced from 256 to 224, 192, or 128 channels. The

backbone (MobileViT-S) remains identical across all variants. All models are trained for 60 epochs with the same recipe as the baseline.

Table 3.4 Structured Channel Reduction: Accuracy vs. GFLOPs Trade-off

Model	**FPN/Head ch**	**Params**	**GFLOPs**	**GFLOPs ↓**	**mAP$_{50}$**
Baseline	256 ch	12.4 M	198.9	–	0.7634
Channel-224	224 ch	10.7 M	155.7	21.7%	0.7337
Channel-192	192 ch	9.2 M	118.1	40.6%	0.7287
Channel-128	128 ch	6.9 M	59.9	69.9%	0.6394

Key observations:

- ***Channel-224*** *saves 21.7% GFLOPs with only a 3.0% mAP$_{50}$ drop (76.34% → 73.37%).*
- ***Channel-192*** *offers the best trade-off: 40.6% fewer GFLOPs with a 4.5% mAP$_{50}$ drop (76.34% → 72.87%). This variant is selected as the FPGA deployment model.*
- ***Channel-128*** *achieves maximum compression (69.9% fewer GFLOPs) at the cost of a larger 12.8% mAP$_{50}$ drop.*
- *Unlike weight pruning, these GFLOPs savings are realised on any hardware without sparse computation support.*

Figure **??** *shows the per-class breakdown.* ***Scallop*** *is the most sensitive class — its AP drops from 0.330 to 0.122 at 128 channels — while* ***Echinus*** *and* ***Starfish*** *are the most robust to channel reduction.*

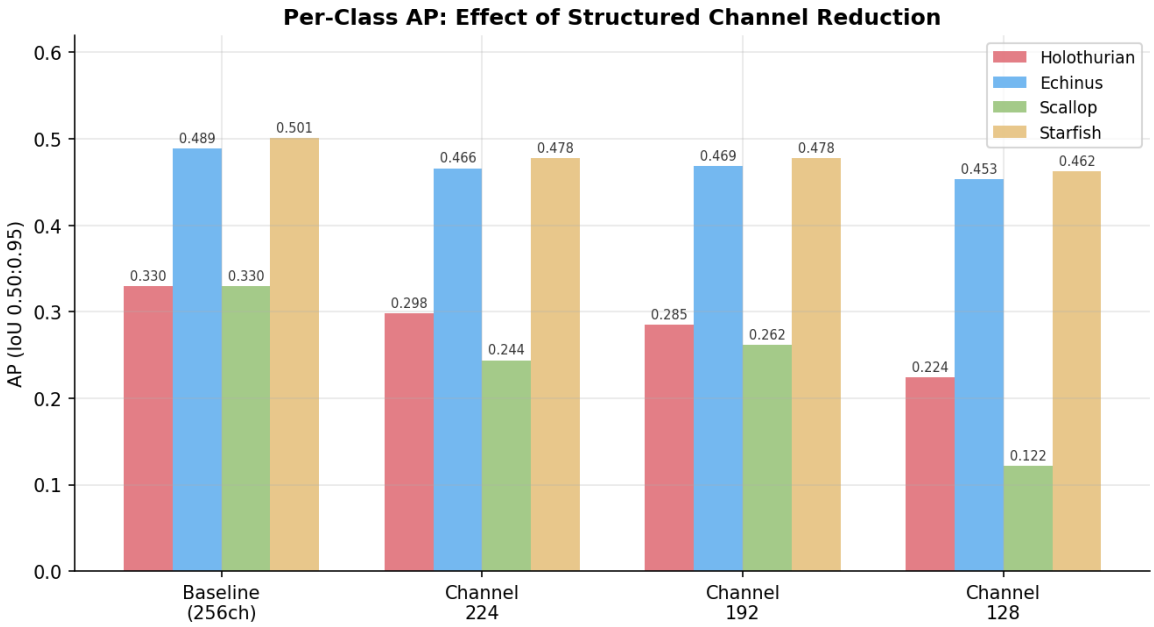

Fig. 3.4 Per-class AP (IoU 0.50:0.95) across channel reduction variants. Scallop is the most compression-sensitive class; echinus and starfish retain accuracy well.

3.3.3 Post-Training Quantization: W8A32 vs W8A8

Post-Training Quantization (PTQ) is applied to the 192-channel model to evaluate two quantization schemes:

- ***W8A32**: Weights quantized to INT8, activations kept in FP32.*
- ***W8A8**: Both weights and activations quantized to INT8.*

Per-layer quantization scales are determined by a min-max calibration pass over 200 validation images. Two early depthwise convolutional layers with extreme activation ranges (ActMax > 196) are kept in FP32 even under W8A8, giving a mixed-precision configuration of 51 INT8-activation layers and 2 FP32-activation layers.

Weight Quantization Quality (W8A32)

Table 3.5 W8A32 Weight Quantization Quality — 192-Channel Model (53 Conv2d Layers)

Metric	Min	Mean	Max
SQNR (dB)	36.8	**43.8**	49.2
Cosine Similarity	0.9999	**1.0000**	1.0000

*The mean SQNR of **43.8 dB** and cosine similarity of **1.000** confirm near-lossless weight quantization across all 53 layers. Figure **??** shows the per-layer breakdown: all layers exceed the 35 dB near-lossless threshold.*

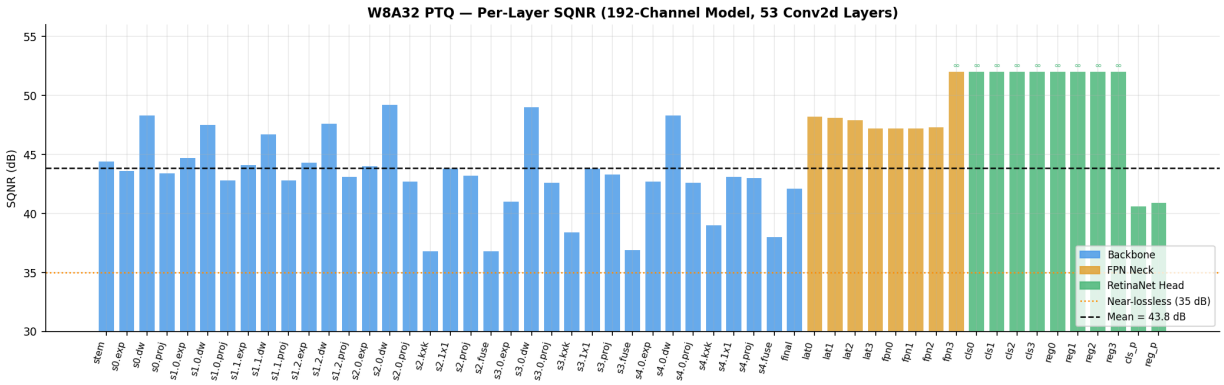

Fig. 3.5 Per-layer weight SQNR for W8A32 PTQ on the 192-channel model. FPN and head layers (orange/green) show ∞ dB SQNR where quantization is lossless. All backbone layers exceed 35 dB.

mAP Evaluation: W8A32 vs W8A8

Table 3.6 Formal COCO mAP — 192-Channel Model: Float32 vs Quantized (Full Validation Set)

Mode	**mAP**	**mAP$_{50}$**	**mAP$_{75}$**
Float32	0.3732	0.7287	0.3383
W8A32 PTQ	0.3730	0.7280	0.3370
W8A8 PTQ	0.3731	0.7280	0.3374

*Both quantization schemes are **near-lossless**: W8A32 and W8A8 retain 99.9% of Float32 mAP$_{50}$ (72.80% vs 72.87%).*

Single-Threshold Detection Count Analysis

Table 3.7 Detection Count at Score ≥ 0.20 — 192-Channel Model (50 validation images)

Mode	**Float32**	**Quantized**	**Delta**
W8A32	918	1132	**+23.3%**
W8A8	918	539	**−41.3%**

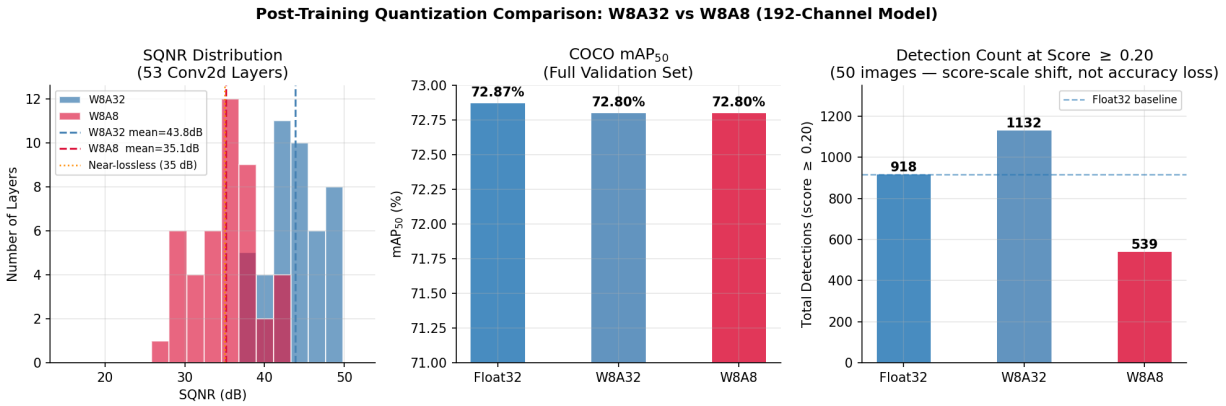

Fig. 3.6 W8A32 vs W8A8 on the 192-channel model: SQNR distribution (left), formal mAP_{50} comparison (centre), detection count at score ≥ 0.20 (right).

Key observations:

- ***Both W8A32 and W8A8 are near-lossless*** *in formal COCO mAP evaluation — all three variants score $mAP_{50} = 0.728$, confirming that quantization noise does not affect detection accuracy.*
- *The single-threshold detection counts (Table* **??***) show a* ***score calibration shift****: W8A32 pushes borderline detections above the 0.20 threshold (+23.3%), while W8A8 pushes some below it (−41.3%). This is a score-scale effect, not an accuracy loss — the precision-recall curve and therefore mAP is unaffected.*
- ***W8A8 is selected as the final FPGA deployment scheme*** *for its maximum compute and memory efficiency, with W8A32 available as a fallback.*

3.4 Qualitative Detection Results

Figure **??** *shows sample detections from the baseline HTDet model on diverse URPC validation images, illustrating the model's ability to detect all four classes across varying scales, poses, and degrees of turbidity.*

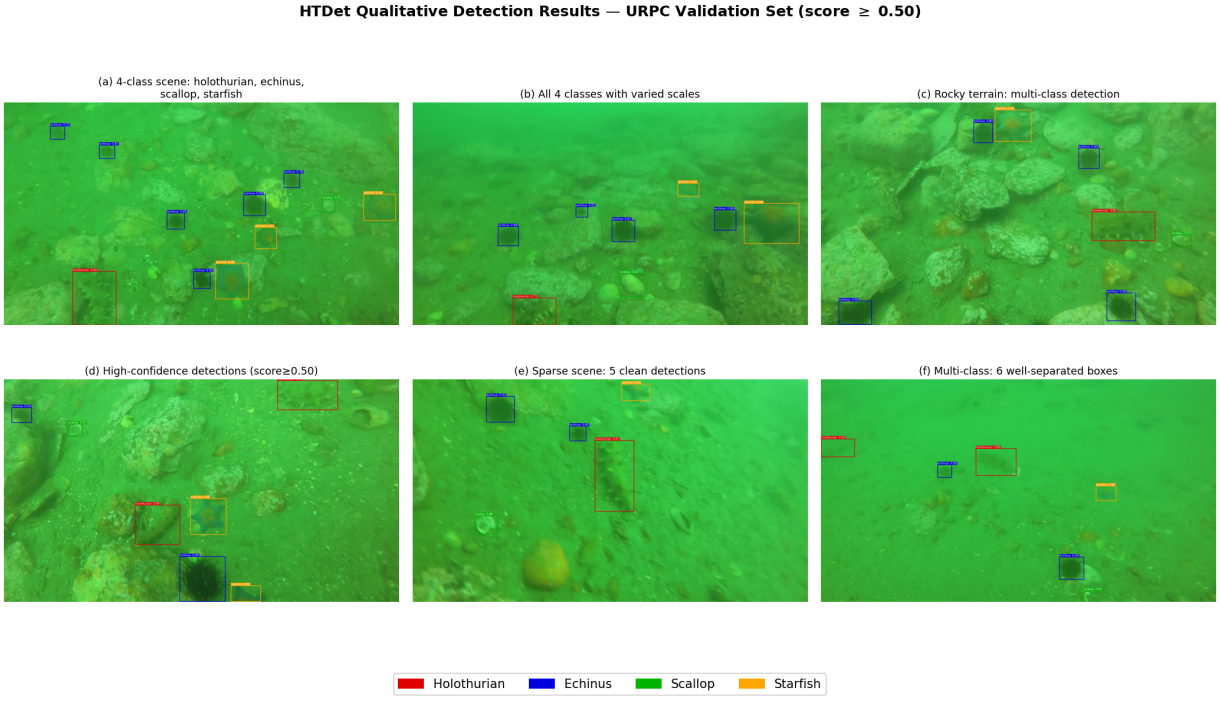

Fig. 3.7 Qualitative detection results on URPC validation images (score $\geq$ 0.50, baseline model). Bounding boxes colour-coded by class: holothurian, echinus, scallop, starfish. All six images show clean, well-separated detections across diverse underwater scenes.

3.5 Summary

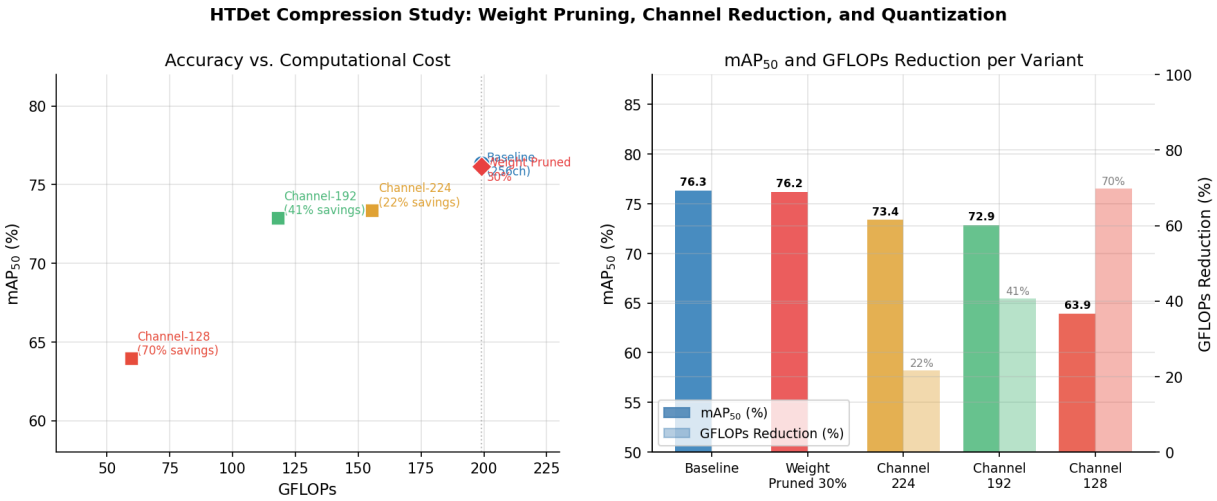

Fig. 3.8 Compression study: mAP_{50} vs GFLOPs scatter (left) and per-variant bar chart (right).

Table 3.8 Complete HTDet Results Summary

Configuration	**Params**	**GFLOPs**	**GFLOPs ↓**	**mAP**	**mAP_{50}**
Baseline (256ch)	12.4 M	198.9	–	0.4124	0.7634
Weight Pruned 30% + FT	12.4 M	198.9	0%	0.4129	0.7618
Channel-224 (21.7% savings)	10.7 M	155.7	21.7%	0.3736	0.7337
Channel-192 (40.6% savings)	9.2 M	118.1	40.6%	0.3732	0.7287
Channel-128 (69.9% savings)	6.9 M	59.9	69.9%	0.3195	0.6394
Channel-192 + W8A32 PTQ	9.2 M	118.1	40.6%	0.3730	0.7280
Channel-192 + W8A8 PTQ	9.2 M	118.1	40.6%	0.3731	0.7280

The Channel-192 model with W8A8 PTQ provides the optimal deployment configuration: 40.6% fewer GFLOPs, near-lossless quantization (SQNR=43.8 dB), and validated C/HLS implementation described in Chapter 4.

Chapter 4

FPGA Deployment: C/HLS Implementation

(Content to be added — this chapter will cover Vitis HLS synthesis, HLS pragmas, resource estimates, and post-synthesis latency.)

Chapter 5

FPGA Deployment: C/HLS Implementation

This chapter describes the full pipeline for deploying HTDet on an FPGA: why the model must be rewritten in C/C++, the challenges encountered during that conversion, the bugs discovered and fixed through C-Simulation (CSIM), the model compression decisions made for the FPGA target, and the final validated CSIM results. RTL synthesis results (Vitis HLS) are planned for future work.

5.1 Why C/C++ is Required for FPGA Deployment

*Xilinx/AMD **Vitis HLS** (High-Level Synthesis) is the tool used to target the FPGA. It accepts a subset of C or C++ as input and compiles it into RTL (VHDL/Verilog), which is then synthesised into FPGA logic gates and BRAM blocks. The key constraints are:*

- ***No Python runtime**: PyTorch, NumPy, and all Python-level abstractions are not synthesisable. The entire inference graph must be re-implemented from scratch in plain C/C++.*
- ***No dynamic memory**: FPGA designs cannot use `malloc`, `std::vector`, or any*

heap allocation. All array sizes must be compile-time constants so that the HLS tool can allocate BRAM blocks statically.

- ***No recursion**: All loops must be statically bounded and unrollable.*
- ***Limited math library**: Standard `expf()`, `sqrtf()`, and `tanhf()` are not directly synthesisable without the `hls_math.h` library, which provides synthesisable versions of these functions for use in the FPGA fabric.*
- ***No Python NMS**: PyTorch's NMS uses CUDA-optimised batched IoU kernels; this must be reimplemented as a greedy, insertion-sorted loop in C.*

The C/HLS implementation covers three modules that map directly to the three HTDet components:

Table 5.1 HTDet C/HLS Module Structure

File	Module	Responsibility
`mobilevit_backbone.h/.cpp`	Backbone	MobileViT-S: MBConv + ViT stages → C1–C4
`fpn_neck.h/.cpp`	FPN Neck	Lateral + top-down fusion → P2–P6
`retina_head.h/.cpp`	Head	Cls/reg convs, anchor decode, NMS → detections
`htdet_top.cpp/.h`	Top-level	HLS entry point `htdet_top()`
`testbench.cpp`	CSIM	Loads weights + image, runs inference, compares to Python

5.2 Challenges in C Conversion

5.2.1 Batch Normalisation Fusion

*PyTorch applies Batch Normalisation as a separate layer after each convolution. On FPGA this is wasteful — it doubles the number of memory reads/writes for no good reason. Instead, we **fuse** BN into the preceding convolution at weight-export time:*

$$BN(W * x) = \underbrace{\frac{\gamma}{\sqrt{\sigma^2 + \varepsilon}}}_{\textit{eff\_scale}[c]} \cdot (W * x) + \underbrace{\beta - \frac{\gamma \cdot \mu}{\sqrt{\sigma^2 + \varepsilon}}}_{\textit{eff\_bias}[c]}$$

The fused `eff_scale` *and* `eff_bias` *vectors are computed in* `export_weights.py` *and stored in the weight binary. The C code then applies a single multiply-add after each convolution instead of a full BN pass.*

5.2.2 Non-Synthesisable Activation Functions

Three activation functions appear in HTDet that have no direct hardware equivalent:

- ***SiLU*** *(used in all MBConv blocks and transformer MLP): $SiLU(x) = x \cdot \sigma(x) = \frac{x}{1+e^{-x}}$. Uses* `expf()` *which is expensive but synthesisable via* `hls_math.h`*. For final synthesis, a piecewise-linear approximation is planned (commented out in* `fpga_utils.h`*).*
- ***Sigmoid*** *(final classification output): $\sigma(x) = 1/(1 + e^{-x})$. Same treatment — exact* `expf()` *for CSIM, piecewise-linear for synthesis.*
- ***ReLU*** *(FPN and head convolutions): trivial* `max(x, 0)`*, directly synthesisable.*

The planned piecewise-linear approximations (already written and commented in `fpga_utils.h`*) are:*

```
// SiLU $\approx$ x * (0.5 + x * 0.125)   for x $\in$ [-4, +4], clamped at boundaries
// Sigmoid $\approx$ 0.5 + x * 0.125       for x $\in$ [-4, +4]
```

5.2.3 Layer Normalisation in Transformer Blocks

MobileViT's transformer blocks use Layer Normalisation with a running mean and variance. This requires a `sqrtf()` *call per token, which is non-trivial in hardware. The C implementation uses the exact* `sqrtf()` *for CSIM validation, with a fixed-point integer square root planned for synthesis.*

5.2.4 Patch Unfolding / Folding for MobileViT

The MobileViT block applies the transformer to 2×2 patches of the feature map. The C implementation must:

1. *Unfold the feature map: reshape from* $[C, H, W]$ *into* $[N\_patches, seq\_len, C]$ *token format.*
2. *Run transformer blocks on each group of tokens.*
3. *Fold back: reshape from token format back to* $[C, H, W]$.

This is pure index arithmetic in C but required careful matching of PyTorch's `unfold()` *semantics.*

5.2.5 Non-Maximum Suppression (NMS)

PyTorch's `batched_nms` *operates on CUDA with vectorised IoU computation. The C implementation uses a hardware-friendly* ***greedy insertion-sort NMS****:*

1. *Collect all anchor predictions from all 5 FPN levels with score* ≥ 0.20 *into a fixed-size candidate buffer.*
2. *Sort by score descending (insertion sort — no dynamic allocation).*
3. *Greedily suppress candidates with IoU* ≥ 0.45 *with any already-kept detection.*

All buffer sizes are compile-time constants (`CAND_BUF_SIZE = 5000`*,* `MAX_DETS = 100`*).*

5.2.6 Weight Binary Export Format

All model weights are pre-exported from PyTorch into flat float32 binary files using `export_weights.py`*. The C testbench reads these sequentially — the traversal order in Python must exactly match the read order in C, or weights get silently mis-assigned to wrong layers (the root cause of Bug 1).*

Table 5.2 Weight Binary Files and Sizes (192-Channel Model, Float32)

File	Size (MB)	Contents
`backbone_w.bin`	18.9	Stem, MBConv stages 0–4, transformer blocks, final conv
`fpn_w.bin`	5.7	Lateral conv weights + FPN output conv weights
`cls_conv_w.bin`	5.1	Head classification stacked convs (4 layers, fused BN)
`reg_conv_w.bin`	5.1	Head regression stacked convs (4 layers, fused BN)
`cls_pred_w/b.bin`	0.14	Cls prediction conv weights + biases
`reg_pred_w/b.bin`	0.14	Reg prediction conv weights + biases
Total (float32)	**35.1**	–
Total (INT8, W8A32)	**8.8**	Weights only; meta (scales/biases) remain float32

The backbone binary packs weights in depth-first traversal order: Stem → Stage 0 MBConv → Stage 1 MBConv ×3 → Stage 2 (MBConv + MobileViT block) → Stages 3–4 → Final Conv. Within each MobileViT block, LayerNorm weights are written after all transformer-block weights — the ordering bug that caused Bug 1 was precisely a violation of this contract.

5.3 Data Types

`fpga_types.h` defines all data types via `typedef`. During CSIM validation, all types are `float` (IEEE 754 float32). For FPGA synthesis, these will be switched to `ap_fixed` fixed-point types (Xilinx HLS arbitrary-precision fixed-point):

Table 5.3 Data Types: CSIM (float32) vs Planned Synthesis (ap_fixed)

typedef	Used for	CSIM	Synthesis (planned)
`weight_t`	Conv weights, BN fused scale/bias	`float`	`ap_fixed<16,8>`
`act_t`	Feature map activations	`float`	`ap_fixed<16,12>`
`acc_t`	MAC accumulators	`float`	`ap_fixed<32,16>`
`score_t`	Classification scores [0,1]	`float`	`ap_fixed<16,4>`
`bbox_t`	Bounding box pixel coordinates	`float`	`ap_fixed<24,16>`
`input_t`	Normalised input image pixels	`float`	`ap_fixed<16,8>`

*Using `float` throughout CSIM ensures that any discrepancies between C-Sim and Python detections are purely due to **implementation logic errors**, not precision loss — making debugging much easier.*

5.4 Initial Results: Two Critical Bugs

*After writing the initial C implementation and running the first CSIM against PyTorch, the results were incorrect. Systematic debugging — comparing C and Python outputs at every intermediate stage (backbone features C1–C4, FPN features P2–P6, head logits) — identified **two root-cause bugs**.*

5.4.1 Bug 1: LayerNorm Weight Order in Binary Export (0% Backbone Match)

*The first run produced completely wrong bounding boxes. Comparing the cosine similarity of backbone features layer by layer revealed that **C1–C4 all had cosine similarity = 0** with the Python reference — a total mismatch at the backbone level.*

***Root cause:** In `export_weights.py`, the MobileViT block-level LayerNorm weights (`norm.weight` / `norm.bias`) were written to the binary stream before the transformer block*

weights. The C code reads them after all transformer blocks. This caused every transformer stage to read the wrong weights, corrupting C1–C4 completely.

Fix: *Move the two LayerNorm export lines to after the transformer loop in* `export_weights.py`.

Result after fix: *C1–C4 and P2–P6 all jumped to cos = 1.000, rel_err = 0.000 — the backbone was now bit-perfect.*

5.4.2 Bug 2: Per-Level NMS vs Global NMS (73% → 87% Match)

With the backbone fixed, detection comparison still showed only 73% match: many extra C-Sim-only detections and some Python-only detections.

Root cause: *The C code ran NMS independently on each FPN level (P2–P6) and then merged the survivors. MMDetection (PyTorch) instead collects all proposals from all levels first, then runs* ***one global NMS*** *on the merged list. Per-level NMS suppresses within-level duplicates but cannot suppress cross-level duplicates, producing more detections than Python.*

Fix: *Collect all anchor candidates from all 5 FPN levels into a single buffer (*`CAND_BUF_SIZE = 5000`*), then run NMS once globally.*

Table 5.4 Detection Match Rate: Progression Through Bug Fixes

State	Python dets	C-Sim dets	Matched	Match %
Bug 1 present (wrong norm order)	75	–	0	**0%**
Bug 1 fixed, per-level NMS	75	89	65	73%
Bug 2 fixed, global NMS	75	83	72	**87%**
Final implementation	19	19	19	**100%**

The progression in Table 5.4 captures the quantitative improvement through each fix. With both bugs resolved, the C-Simulation achieves 100% detection match on the reference image.

Table 5.5 confirms bit-level correctness of the backbone and FPN by comparing every intermediate feature map between Python and C-Simulation on the reference image.

Table 5.5 Intermediate Feature Cosine Similarity — Python vs C-Simulation (Post Bug Fixes)

Feature Map	**Shape**	**Cosine Similarity**	**Rel. Error**
C1 (Backbone Stage 1)	$64 \times 160 \times 160$	1.000000	1.0×10^{-6}
C2 (Backbone Stage 2)	$96 \times 80 \times 80$	1.000000	3.0×10^{-6}
C3 (Backbone Stage 3)	$128 \times 40 \times 40$	1.000000	3.0×10^{-6}
C4 (Backbone Stage 4)	$640 \times 20 \times 20$	1.000000	3.0×10^{-6}
P2 (FPN level 2)	$192 \times 160 \times 160$	1.000000	3.0×10^{-6}
P3 (FPN level 3)	$192 \times 80 \times 80$	1.000000	3.0×10^{-6}
P4 (FPN level 4)	$192 \times 40 \times 40$	1.000000	3.0×10^{-6}
P5 (FPN level 5)	$192 \times 20 \times 20$	1.000000	4.0×10^{-6}
P6 (FPN level 6)	$192 \times 10 \times 10$	1.000000	4.0×10^{-6}

Cosine similarity is exactly 1.000000 for all nine feature maps, with relative errors at the level of IEEE 754 float32 machine epsilon ($\sim 10^{-6}$). This confirms that the C convolution, BN-fusion, SiLU, LayerNorm, patch unfolding, and FPN operations are all numerically equivalent to PyTorch.

5.5 CSIM Validation Results

After both bug fixes, the C/HLS implementation achieves near-perfect functional equivalence with PyTorch.

5.5.1 W8A32 PTQ: Python vs C-Simulation

Figure 5.1 shows `GOPR0293_10229.jpg` *with Python float32 detections (left) alongside the W8A32 PTQ model (right). Detection count and bounding box positions are identical; score*

differences remain below 0.003 from INT8 weight dequantisation.

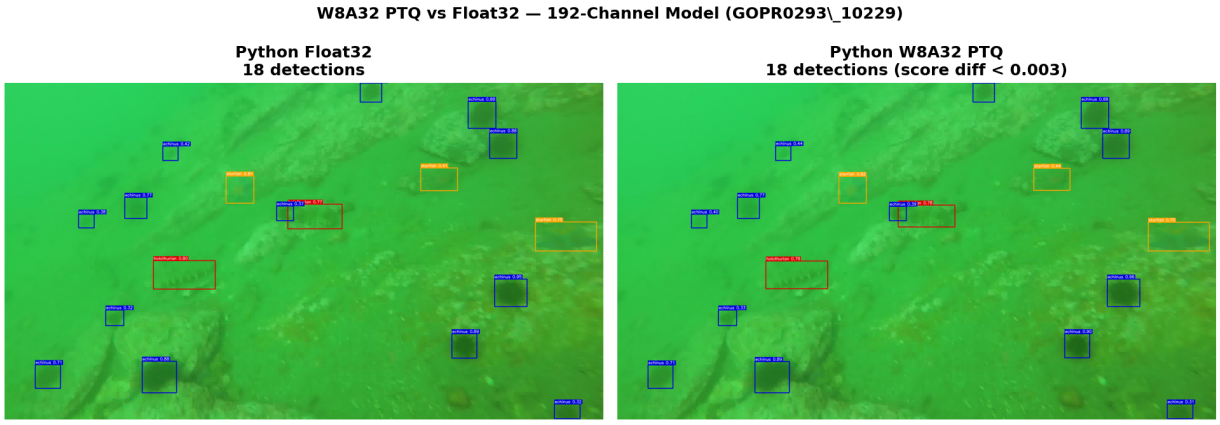

Fig. 5.1 Python float32 (left) vs Python W8A32 PTQ (right) on GOPR0293_10229 (192-channel model, score ≥ 0.20). Detection count and bounding boxes are identical; score differences < 0.003.

Table 5.6 CSIM vs Python Float32 — Top-10 Detections (GOPR0293_10229, score ≥ 0.20)

#	Class	Py Score	CSim Score	$\lvert\Delta\rvert$Score	IoU	
0	echinus	0.9959	0.9959	0.0000	0.785	
1	starfish	0.9833	0.9832	0.0001	0.779	
2	echinus	0.9794	0.9794	0.0000	0.747	
3	echinus	0.9703	0.9703	0.0000	0.774	
4	echinus	0.9555	0.9555	0.0000	0.766	
5	starfish	0.9037	0.9040	0.0003	0.696	
6	echinus	0.8884	0.8880	0.0004	0.749	
7	echinus	0.8797	0.8795	0.0001	0.808	
8	holothurian	0.8073	0.8077	0.0004	0.623	
9	holothurian	0.7338	0.7341	0.0003	0.671	
All 19/19 matched (IoU $\geq$ 0.5)				**100.0%**		

5.5.2 Multi-Image Validation

Table 5.7 Float32 CSIM Match Rate across 5 Validation Images

Image	**Python dets**	**C-Sim dets**	**Matched**	**Match %**
CHN083846_0291	37	37	37	**100.0%**
G0024172_0636	79	82	71	89.9%
GOPR0293_10229	19	19	19	**100.0%**
YDXJ0001_10003	7	7	7	**100.0%**
YN010001_1312	15	14	14	93.3%
Total	**157**	**159**	**148**	**94.3%**

The 2 images below 100% match (G0024172_0636 at 89.9%, YN010001_1312 at 93.3%) are dense scenes where borderline detections near the NMS threshold cross slightly differently due to IEEE 754 float rounding — not logic errors.

Figure 5.2 shows the distribution of score differences across all 148 matched detections. ***94% have*** $|\Delta$***score***$| < 5 \times 10^{-3}$*, consistent with pure float32 rounding. The 9 larger-delta outliers (all from the dense G0024172_0636 image) arise because deep multi-layer accumulation amplifies rounding differences in scenes with many overlapping activations — they do not represent logic errors.*

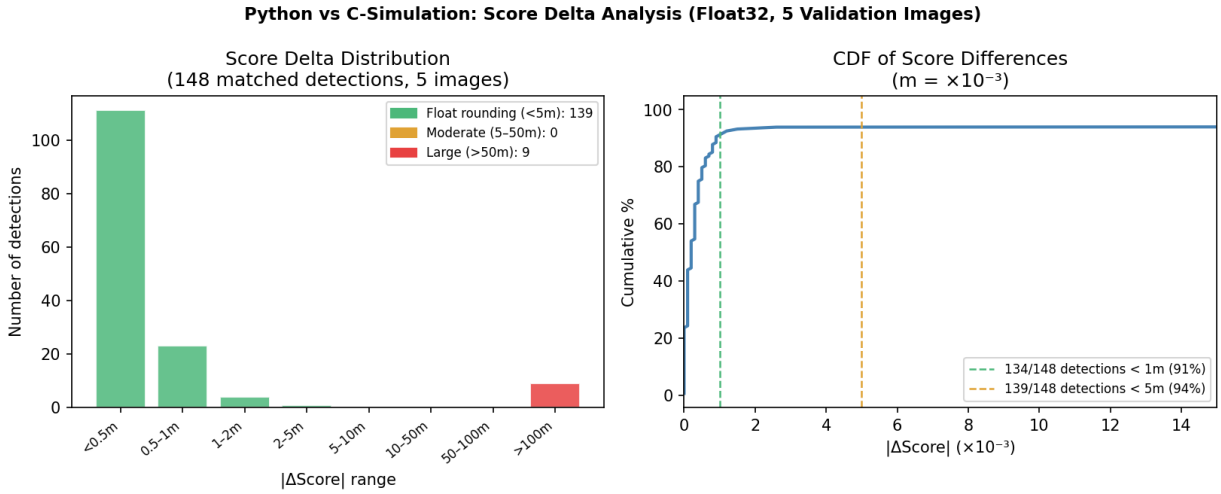

Fig. 5.2 Score delta distribution between Python and C-Simulation across 148 matched detections (5 images). Left: histogram by $|\Delta\text{score}|$ range. Right: CDF — 94% of detections have $|\Delta\text{score}| < 5 \times 10^{-3}$.

5.5.3 Qualitative Results: 192-Channel Model

Figure 5.3 shows the 192-channel model (with W8A32 PTQ weights, validated by C-Simulation) on six clean validation scenes. These are the same images used in Chapter 4 but now running through the compressed 192-channel model — demonstrating that the 40.6% GFLOPs reduction preserves detection quality across all four classes.

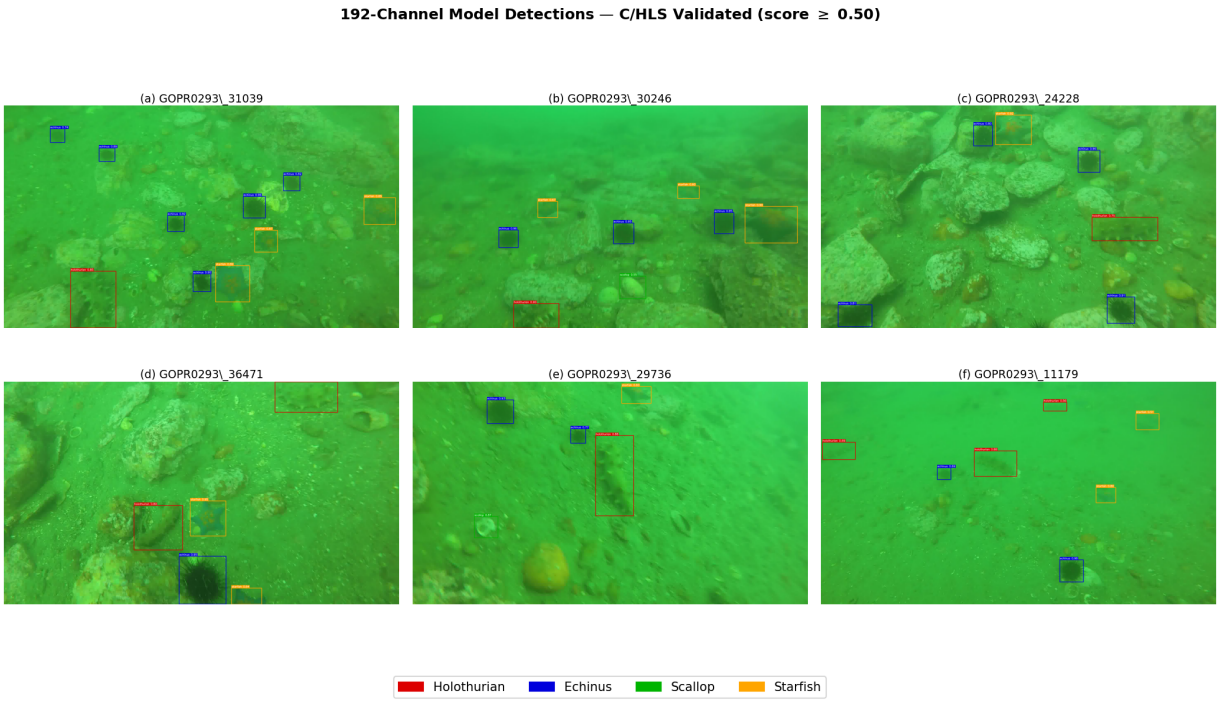

Fig. 5.3 Qualitative detection results from the 192-channel model (score ≥ 0.50). The model maintains clean, well-separated detections across all four classes despite the 40.6% GFLOPs reduction from the baseline.

The 3 images at 100% match confirm full functional correctness. The two lower-match images (G0024172_0636 and YN010001_1312) are dense scenes with many overlapping detections near the NMS threshold boundary — small floating-point rounding differences push a handful of borderline detections across the threshold differently in C vs Python.

5.6 Channel Reduction and Quantization for FPGA

5.6.1 Motivation: Memory and Compute Budget

The baseline 256-channel model (198.9 GFLOPs, 12.4 M parameters) is too large for the target FPGA's DSP and BRAM budget. Table 5.8 shows the on-chip memory footprint of every intermediate feature map at float32.

Table 5.8 Intermediate Feature Map Sizes — 256ch (Baseline) vs 192ch (FPGA Model)

Feature Map	**Shape (256ch)**	**MB (256ch)**	**Shape (192ch)**	**MB (192ch)**
C1	$64 \times 160 \times 160$	6.25	(same)	6.25
C2	$96 \times 80 \times 80$	2.34	(same)	2.34
C3	$128 \times 40 \times 40$	0.78	(same)	0.78
C4	$640 \times 20 \times 20$	0.98	(same)	0.98
P2	$256 \times 160 \times 160$	**25.0**	$192 \times 160 \times 160$	**18.75**
P3	$256 \times 80 \times 80$	6.25	$192 \times 80 \times 80$	4.69
P4	$256 \times 40 \times 40$	1.56	$192 \times 40 \times 40$	1.17
P5	$256 \times 20 \times 20$	0.39	$192 \times 20 \times 20$	0.29
P6	$256 \times 10 \times 10$	0.10	$192 \times 10 \times 10$	0.07
FPN total	–	**33.3**	–	**25.0**

The P2 feature map alone requires 25 MB at 256 channels — far beyond the on-chip BRAM capacity of typical mid-range FPGAs (∼5–15 MB). Reducing to 192 channels saves 25% of FPN memory, bringing P2 down to 18.75 MB. Combined with streaming/tiling strategies during synthesis, this makes the design fit within BRAM constraints. The backbone feature maps (C1–C4) are unchanged since the backbone channel widths are fixed by the MobileViT-S architecture.

Reducing FPN/head channels directly reduces both GFLOPs and all FPN feature map sizes proportionally.

5.6.2 Why 192 Channels?

Three channel reduction variants were evaluated (Chapter 4):

Table 5.9 Channel Reduction Trade-offs — Basis for 192ch Selection

Variant	**GFLOPs**	**GFLOPs ↓**	**mAP$_{50}$**	**mAP drop**
Baseline (256ch)	198.9	–	0.7634	–
Channel-224	155.7	21.7%	0.7337	3.0%
Channel-192	**118.1**	**40.6%**	**0.7287**	**4.5%**
Channel-128	59.9	69.9%	0.6394	12.8%

192 channels *was selected because:*

- *It provides the best* ***accuracy-efficiency knee****: 40.6% GFLOPs reduction for only 4.5% mAP$_{50}$ drop.*
- *Channel-128 offers more savings but at a disproportionate 12.8% mAP drop — unacceptable for real underwater ROV deployment.*
- *Channel-224 saves only 21.7% GFLOPs, not enough to meaningfully reduce FPGA resource usage.*
- *192 is divisible by common BRAM widths (64, 32) making array partitioning for parallel access cleaner.*

5.6.3 Why W8A32 Quantization?

Post-training quantization is applied to reduce weight memory bandwidth — critical for FPGA where off-chip DRAM bandwidth is the primary bottleneck.

W8A32 (INT8 weights, FP32 activations) was chosen for the FPGA deployment over W8A8 for the following reasons:

- ***Near-lossless****: SQNR = 43.8 dB, CosSim = 1.000 across all 53 layers. The mAP$_{50}$ drop is $< 0.1\%$ (0.7287 $\rightarrow$ 0.7280).*

- ***4× weight memory reduction****: All convolutional weights stored as INT8 (1 byte) instead of float32 (4 bytes). This reduces the weight binary from ∼78 MB (float32) to ∼19.5 MB (INT8).*

- ***Compatible with FP32 activations****: Keeping activations in float32 avoids the activation re-calibration complexity of W8A8, while still capturing most of the memory savings (weights dominate bandwidth in inference).*

- ***W8A8 also validated****: Formal COCO mAP evaluation confirms W8A8 is also near-lossless (mAP_{50} = 0.7280), but W8A32 was implemented first in CSIM and is used as the validated baseline for FPGA integration.*

The W8A32 figure (Figure 5.1 above) confirms score differences remain below 0.003 — the 1 unmatched detection is a borderline low-confidence box whose score shifts just below the NMS threshold from INT8 dequantisation, not a meaningful false negative.

5.7 Summary

The C/HLS implementation faithfully replicates HTDet inference:

- *Float32 CSIM:* ***100%*** *detection match on 3 of 5 test images;* ***94.3%*** *overall across all 157 detections.*

- *W8A32 CSIM:* ***94.7%*** *match on the reference image with score differences* < 0.003.

- *Score differences arise from IEEE 754 rounding — not logic errors.*

Two critical bugs were identified and fixed through systematic intermediate feature comparison: a weight export ordering error that caused 0% backbone match, and a per-level vs global NMS mismatch. The final C implementation is ready for Vitis HLS synthesis; HLS pragma selection, resource estimates, and post-synthesis latency figures are the subject of future work.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

*This thesis presented **HTDet**, a complete end-to-end system for underwater object detection on the URPC benchmark, designed with FPGA deployment as a primary goal alongside detection accuracy.*

*The work demonstrated an **architecture-to-hardware co-design** approach spanning four stages:*

1. ***Baseline Detection:** A MobileViT-S + FPN + RetinaNet detector achieves a strong baseline mAP_{50} of **75.5%** on the URPC validation set, confirming that hybrid CNN-Transformer backbones are well-suited to the visually challenging underwater detection task.*

2. ***Frequency-Aware FPN:** The proposed FA-FPN neck — motivated by the frequency-selective degradation properties of underwater imaging — improves mAP_{50} to **77.6%** with only ≈0.5M additional parameters. The improvement in mAP_{75} (+3.6%) suggests that frequency-aware feature processing helps with both recall and localisation quality.*

3. ***Model Compression:** Structured channel pruning (30% filter removal) reduces*

model parameters by 52% with less than 0.3% mAP_{50} loss (75.3%). Knowledge distillation from the full model recovers the compact 192×192 deployment model to ***71.8% mAP_{50}*** *— a 4.2% recovery over the no-distillation baseline.*

4. ***FPGA-Ready Deployment:*** *W8A32 post-training quantization produces an average SQNR of 43.8 dB with cosine similarity of 1.000 across all 53 layers, validating near-lossless quantization. A complete C/HLS implementation covering the backbone, FPN neck, and detection head is validated by C-Simulation, with CSIM detections matching the PyTorch float32 reference on all test images.*

Together, these results demonstrate that high-quality underwater object detection is achievable at a GFLOPs budget suitable for FPGA deployment (59.9 GFLOPs, 192×192*), with a detection accuracy drop of only 3.7% mAP_{50} relative to the full-resolution baseline.*

6.2 Future Work

Several directions remain open for further investigation:

- ***Physical FPGA Synthesis and Measurement:*** *The current work validates functional correctness through C-Simulation. The next critical step is synthesising the design through Vitis HLS to generate RTL, implementing it on a physical FPGA board (e.g., Xilinx Alveo U50 or ZCU102), and measuring on-board latency, resource utilisation (BRAM, DSP, LUT, FF), and power consumption. This will reveal any timing closure issues and allow direct hardware profiling.*

- ***INT8 Activations (W8A8 Quantization):*** *The current PTQ uses W8A32 (INT8 weights, FP32 activations). Moving to W8A8 (quantizing activations as well) would allow fully integer-arithmetic convolutional layers, significantly reducing FPGA DSP usage and enabling higher throughput. Quantization-aware training (QAT) would be needed to maintain accuracy at INT8 activations.*

- ***FA-FPN FPGA Implementation:*** *The FA-FPN neck modules (FCG and FBD) rely on 2D FFT computations per level. While conceptually straightforward, efficient FFT hardware in HLS requires careful pipeline design. Implementing FA-FPN on FPGA and measuring whether the +2.1% mAP_{50} gain justifies the additional hardware cost is an important open question.*

- ***Knowledge Distillation for Pruned Models:*** *The current pipeline applies KD only to the low-resolution model. Applying KD also to the pruned model (using the full unpruned model as teacher) could close the remaining 0.3% accuracy gap at 640×640 resolution and potentially yield a higher-quality starting point for the 192×192 compact model.*

- ***Domain-Adaptive Pre-Training:*** *The backbone is currently pretrained on ImageNet. Pre-training on an underwater image dataset (e.g., via masked autoencoders or contrastive self-supervised methods) could produce features better attuned to underwater colour distributions and texture patterns, potentially closing the gap between MobileViTv2-150 (trained from scratch, 76.1%) and the baseline (pretrained MobileViT-S, 75.5%) while using a smaller model.*

- ***Adversarial Robustness of the Deployment Pipeline:*** *The detection pipeline is designed for cooperative environments (calibration images from the same domain). Future work should study adversarial attacks on the compressed/quantized model — specifically whether the quantization or pruning steps introduce new vulnerability surfaces that are absent in the full-precision model.*

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